Problem 1

Probability to have N_s photons in a state $s = (\vec{k}, z)$, where $z = \pm 1$ (for two polarizations), is

$$w_s(N_s) = \frac{e^{-\frac{\hbar\omega_s}{k_B T}\left(N_s + \frac{1}{2}\right)}}{Z}$$

Partition function is found from normalization condition:

$$Z = \sum_{N_s=0}^{\infty} e^{-\frac{\hbar\omega_s}{k_B T}\left(N + \frac{1}{2}\right)} = \frac{e^{-\frac{x}{2}}}{1 - e^{-x}}, \text{ where } x = \frac{\hbar\omega_s}{k_B T}$$

Thus, $$w_s(N_s) = e^{-N_s x}\left(1 - e^{-x}\right)$$

Average number of photons in the state 's' is

$$\overline{N_s} = \sum_{N_s=0}^{\infty} N_s w_s(N_s) = \left(1 - e^{-x}\right) \sum_{N_s=0}^{\infty} N_s e^{-N_s x} =$$

$$= -\left(1 - e^{-x}\right) \frac{\partial}{\partial x} \sum_{N_s=0}^{\infty} e^{-N_s x} = -\left(1 - e^{-x}\right) \frac{\partial}{\partial x} \frac{1}{1 - e^{-x}} =$$

$$= \left(1 - e^{-x}\right) \frac{e^{-x}}{\left(1 - e^{-x}\right)^2} = \frac{1}{e^{x} - 1} = \frac{1}{e^{\frac{\hbar\omega_s}{k_B T}} - 1}$$

This is Bose-Einstein distribution with zero chemical potential.

Average energy in state s is $E_s = \hbar\omega_s \overline{N_s}$, where $\omega_s = c|\vec{k}|$

The total energy (per unit volume) is found by multiplying by the number of states per element of k-space:

$$\frac{E}{V} = 2\int \frac{d^3k}{(2\pi)^3} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1}$$

to account for polarization degeneracy

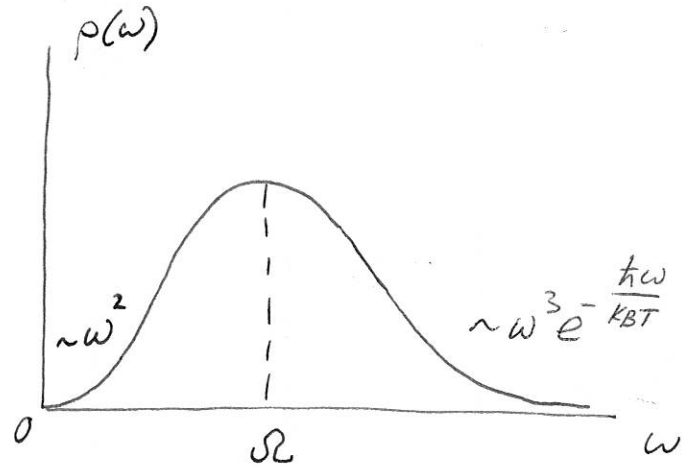
using $d^3k = 4\pi k^2 dk = 4\pi\omega^2 d\omega / c^3$ we write:

$$\frac{E}{V} = \frac{\hbar}{\pi^2 c^3} \int_0^\infty \frac{\omega^3 d\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1}$$

This can be re-written as $\frac{E}{V} = \int_0^\infty \rho(\omega) d\omega$ where spectral energy density

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\frac{\hbar\omega}{k_BT}} - 1}$$

has the form



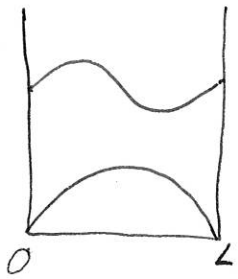
The of $\rho(\omega)$ is found from

$$\left(\frac{x^3}{e^x - 1}\right)' = \frac{3x^2(e^x - 1) - x^3 e^x}{(e^x - 1)^2} = 0 \Rightarrow 3(e^x - 1) = xe^x$$

$x = 2{,}82$

$$\Omega = \frac{k_BT}{\hbar} \times 2{,}82$$

Problem 2



$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi n x}{L}$$

$$\psi_m(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi m x}{L}$$

$$d_{nm} = \frac{2q}{L} \int_0^L dx\, x \sin \frac{\pi n x}{L} \sin \frac{\pi m x}{L} \quad , n \neq m$$

using $\sin a \sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$

we can write

$$d_{nm} = \frac{q}{L} \int_0^L dx\, x \left[\cos \frac{\pi(n-m)x}{L} - \cos \frac{\pi(n+m)x}{L}\right] \qquad (1)$$

the first integral:

$$\frac{1}{L} \int_0^L dx\, x \cos \frac{\pi(n-m)x}{L} = \frac{x}{\pi(n-m)} \sin \frac{\pi(n-m)x}{L} \Big|_0^L$$

$$- \frac{1}{\pi(n-m)} \int_0^L dx \sin \frac{\pi(n-m)x}{L} =$$

$$= \frac{L}{\pi^2(n-m)^2} \cos \frac{\pi(n-m)x}{L} \Big|_0^L = -\frac{L}{\pi^2(n-m)^2}\left(1 - (-1)^{n-m}\right)$$

Similarly, the second integral in Eq. (1) is obtained by replacing $m \to -m$:

$$d_{nm} = q \frac{L}{\pi^2}\left[\frac{1-(-1)^{n+m}}{(n+m)^2} - \frac{1-(-1)^{n-m}}{(n-m)^2}\right]$$

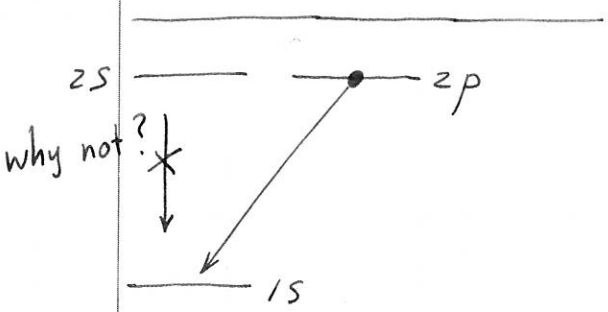
This expression vanishes if $\Delta n = n - m$ is even ($n+m$ is then even too) $\longrightarrow$ only transitions between states of opposite parity are allowed, since operator $\vec{d} = q\vec{r}$ is odd under spatial inversion (with respect to $x = L/2$).

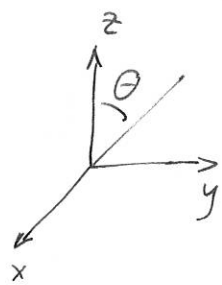
Problem 3

The rate of transition is given by the equation (see Feb. 12 lecture notes)

$$w = \frac{\omega^3 |\vec{d}_{fi}|^2}{3\pi c^3 \varepsilon_0 \hbar}, \quad \text{where} \quad \hbar\omega = \frac{me^4}{2(4\pi\varepsilon_0)^2\hbar^2}\left(1 - \frac{1}{4}\right) =$$

$$= \frac{3}{8}\frac{me^4}{(4\pi\varepsilon_0\hbar)^2}$$





Only d_z operator has non-zero matrix element for the $2p \to 1s$ transition

$$(d_z)_{fi} = e\int 2\pi r^2 dr \sin\theta d\theta \cdot r\cos\theta \; \psi_{2p}(r,\theta)\, \psi_{1s}(r)$$

$$= \frac{2\pi e \cdot 2}{a_B^{3/2} \cdot 2\sqrt{6}\, a_B^{3/2}} \sqrt{\frac{1}{4\pi}} \sqrt{\frac{3}{4\pi}} \int_0^\infty r^3 dr \left(\frac{r}{a_B}\right) e^{-\frac{3r}{2a_B}} \times$$

$$\times \int_0^\pi d\theta \sin\theta \cos^2\theta$$

The integrals here

$$\int_0^\infty r^3 dr \frac{r}{a_B} e^{-\frac{3r}{2a_B}} = a_B^4 \int_0^\infty dz\, z^4 e^{-\frac{3z}{2}} = \frac{256}{81} a_B^4$$

$$\int_0^\pi d\theta \sin\theta \cos^2\theta = \int_{-1}^{1} dy\, y^2 = \frac{2}{3}$$

All together:

$$(d_z)_{fi} = \frac{2\pi e a_B}{\sqrt{2}\, 4\pi} \times \frac{256}{81} \times \frac{2}{3} = \frac{128\sqrt{2}}{243}\, e a_B$$

All together:

$$w = \frac{1}{3\pi c^3 \varepsilon_0 \hbar} \left(\frac{3}{8\hbar} \underbrace{\frac{m e^4}{(4\pi\varepsilon_0 \hbar)^2}}_{\frac{\hbar^2}{m a_B^2}} \right)^3 \left(\frac{128\sqrt{2}}{243} \right)^2 e^2 a_B^2$$

also $\frac{e^2}{4\pi\varepsilon_0 c\hbar} = \alpha = \frac{1}{137}$

$$w = \frac{4}{3} \frac{\alpha a_B^2}{c^2} \left(\frac{3}{8}\right)^3 \frac{\hbar^3}{m^3 a_B^6} \left(\frac{128\sqrt{2}}{243} \right)^2 =$$

$$= \frac{256}{6561}\, \alpha \frac{\hbar^3}{c^2 m^3 a_B^4}$$